Line Item 0001:
Description: "Thin" Si and SiGe layers on Si – Spec 1
Quantity: 15 wafers

1. Substrate
   1.1. Prime double side polished (DSP) silicon
   1.2. Orientation: <100> with SEMI standard primary wafer flat
   1.3. Thickness: 525 μm ± 25 μm
   1.4. Dopant: phosphorus, resistivity range 1 **Ω**·cm to 100 **Ω**·cm
2. Film Stack per table below

Layer	Thickness [nm]	Composition [at]	Dopant [at·cm^{-3}]
7	10 ± 2	Si	B, 2×10^{14} to 5×10^{15}
6	10 ± 2	$Si_{0.7}Ge_{0.3}$	B, 5×10^{19} to 5×10^{20}
5	10 ± 2	Si	B, 2×10^{14} to 5×10^{15}
4	10 ± 2	$Si_{0.7}Ge_{0.3}$	B, 5×10^{19} to 5×10^{20}
3	10 ± 2	Si	B, 2×10^{14} to 5×10^{15}
2	10 ± 2	$Si_{0.7}Ge_{0.3}$	B, 5×10^{19} to 5×10^{20}
1	10 ± 2	Si	B, 2×10^{14} to 5×10^{15}
Substrate			

Line Item 0002:
Description: "Thin" Si and SiGe layers on Si – Spec 2
Quantity: 15 wafers

1. Substrate
   1.1. Prime double side polished (DSP) silicon
   1.2. Orientation: <100> with SEMI standard primary wafer flat
   1.3. Thickness: 525 μm ± 25 μm
   1.4. Dopant: phosphorus, resistivity range 1 **Ω**·cm to 100 **Ω**·cm
2. Film Stack per table below

Layer	Thickness [nm]	Composition [at frac]	Dopant [at·cm^{-3}]
7	25 ± 5	Si	
6	20 ± 5	$Si_{0.55}Ge_{0.45}$	
5	25 ± 5	Si	
4	30 ± 5	$Si_{0.70}Ge_{0.30}$	B, 2×10^{14} to 5×10^{15}
3	25 ± 5	Si	
2	40 ± 5	$Si_{0.85}Ge_{0.15}$	
1	25 ± 5	Si	
Substrate			

Line Item 0003: OPTION LINE ITEM 1
Description: "Thin" doped Si on Si
Quantity: At least 10 wafers, lot size specified at time of order

1. Fixed Parameters
   1.1. Substrate: 100 mm prime double side polished <100> Si wafers, 525 μm thick, SEMI standard flats.
   1.2. Film stack: Fully coherent Si with B and Ph dopants at concentrations up to 5×10^{20} cm^{-3} (atomic, no requirement for electrically active). Total film stack thickness up to 1 μm with up to 5 layers, with no layers thinner than 10 nm.
   1.3. Minimum lot size: 10 wafers
2. Parameters configurable at time of Modification
   2.1. Substrate resistivity: The dopant (B, N, Ph, As, and Sb) and target resistivity (within ±10×)

2.2. Film stack: The thickness and doping of each layer
2.3. Lot size: Minimum order quantity if 10 wafers, but additional wafers of the same lot may be ordered with an additional per wafer cost.

Line Item 0004: OPTION LINE ITEM 2
Description: "Thick" doped Si on Si
Quantity: At least 10 wafers, lot size specified at time of Modification

1. Substrate: 100 mm prime double side polished <100> Si wafers, 525 μm thick, SEMI standard flats.
   1.1. Film stack: Fully coherent Si with B and Ph dopants at concentrations up to 5×10^{20} cm^{-3}. Total film thickness up to 10 μm in one layer.
   1.2. Minimum lot size: 10 wafers
2. Parameters configurable at time of order
   2.1. Substrate resistivity: The dopant (B, N, Ph, As, and Sb) and target resistivity (within ±10×)
   2.2. Film stack: The thickness and doping of the doped Si layer
   2.3. Lot size: Minimum order quantity if 10 wafers, but additional wafers of the same lot may be ordered with an additional per wafer cost.

Line Item 0005: OPTION LINE ITEM 3
Description: Isotopically enriched ^{28}Si on Si
Quantity: At least 10 wafers, lot size specified at time of Modification

1. Fixed Parameters
   1.1. Substrate: 100 mm prime double side polished <100> Si wafers, 525 μm thick, SEMI standard flats.
   1.2. Film stack: Fully coherent Si or ^{28}Si with B, N, and Ph dopants at concentrations up to 5×10^{20} cm^{-3}. Total film thickness up to 100 nm with no layers thinner than 5 nm.
   1.3. Minimum lot size: 10 wafers
2. Parameters configurable at time of order
   2.1. Substrate resistivity: The dopant (B, N, Ph, As, and Sb) and target resistivity (within ±10×)
   2.2. Film stack: The thickness, doping, and isotopic composition of each layer.

Line Item 0006: OPTION LINE ITEM 4
Description: "Thin" Si and SiGe layers on Si
Quantity: At least 10 wafers, lot size specified at time of Modification

1. Fixed Parameters
   1.1. Substrate: 100 mm prime double side polished <100> Si wafers, 525 μm thick, SEMI standard flats.
   1.2. Film stack: Fully coherent Si and $Si_{1-x}Ge_x$ superlattices where x is between 0.1 and 0.4. Total film thickness up to 250 nm with no layer thinner than 5 nm. B and Ph dopants at concentrations up to 5×10^{20} cm^{-3}.
   1.3. Minimum lot size: 10 wafers
2. Parameters configurable at time of order
   2.1. Substrate resistivity: The dopant (B, N, Ph, As, or Sb) and target resistivity (within ±10×)
   2.2. Film stack: The thickness, Ge composition, and doping of each layer
   2.3. Lot size: Minimum order quantity if 10 wafers, but additional wafers of the same lot may be ordered with an additional per wafer cost.

Line Item 0007: OPTION LINE ITEM 5
Description: "Thick" SiGe on buffered Si
Quantity: At least 10 wafers, lot size specified at time of Modification

1. Fixed Parameters
   1.1. Substrate: 100 mm prime double side polished <100> Si wafers, 525 μm thick, SEMI standard flats.
   1.2. Film stack: Virtual substrate created by relaxed buffer layer(s) of SiGe with a fully relaxed (relative to Si) $Si_{1-x}Ge_x$ layer on top where x is between 0.1 and 0.4. Thickness of the relaxed SiGe film thickness less than 10 μm.
   1.3. Minimum lot size: 10 wafers

2. Parameters configurable at time of order
   2.1. Substrate resistivity: The dopant (B, N, Ph, As, or Sb) and target resistivity (within ±10×)
   2.2. Film stack: Relaxed SiGe thickness, Ge composition, and doping of the relaxed SiGe layer. Vendors may create a virtual substrate with layers of SiGe buffers that matches the needs of the specified relaxed SiGe layer.
   2.3. Lot size: Minimum order quantity if 10 wafers, but additional wafers of the same lot may be ordered with an additional per wafer cost.

I. SCHEDULE OF DELIVERABLES

Define all deliverables and indicate (1) the quantity desired; (2) the format in which each deliverable is to be supplied; (3) the delivery date (based on the number of weeks/days); (4) the place of delivery. The context of each deliverable shall be clear.

Deliverable Number	***Description***	***Quantity***	***Due Date***	***Place of Delivery***
1	CLIN 001 CLIN 002	15 wafers 15 wafers	Within 20 weeks of award	NIST Gaithersburg
2	Future Orders		Within 20 weeks of option modification	NIST Gaithersburg